



Harshitha S 250701234-2025-CSE ▾

**H2**

|              |                                 |
|--------------|---------------------------------|
| Started on   | Friday, 21 August 2026, 4:09 PM |
| State        | Finished                        |
| Completed on | Friday, 21 August 2026, 4:11 PM |
| Time taken   | 1 min 37 secs                   |
| Marks        | 1.00/1.00                       |

**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int binarySearch(int a[], int low, int high, int x)
4  {
5      if (low > high)
6          return -1;
7
8      int mid = (low + high) / 2;
9
10     if (a[mid] == x)
11         return mid;
12     else if (a[mid] < x)
13         return binarySearch(a, mid + 1, high, x);
14     else
15         return binarySearch(a, low, mid - 1, x);
16 }
17
18 int main()
19 {
20     int n, x;
21     scanf("%d", &n);
22
23     int a[n];
24
25     for (int i = 0; i < n; i++)
26         scanf("%d", &a[i]);
27
28     scanf("%d", &x);
29
30     for (int i = 0; i < n; i++)
31     {
32         int need = x - a[i];
33
34         int pos = binarySearch(a, i + 1, n - 1, need);
35
36         if (pos != -1)
37         {
38             printf("%d\n", a[i]);
39             printf("%d\n", a[pos]);
40             return 0;
41         }
42     }
43
44     printf("No\n");
45
46     return 0;
47 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 4        | 4   | ✓ |
|   | 2     | 10       | 10  |   |
|   | 4     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 14    |          |     |   |
| ✓ | 5     | No       | No  | ✓ |
|   | 2     |          |     |   |
|   | 4     |          |     |   |
|   | 6     |          |     |   |
|   | 8     |          |     |   |
|   | 10    |          |     |   |
|   | 100   |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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